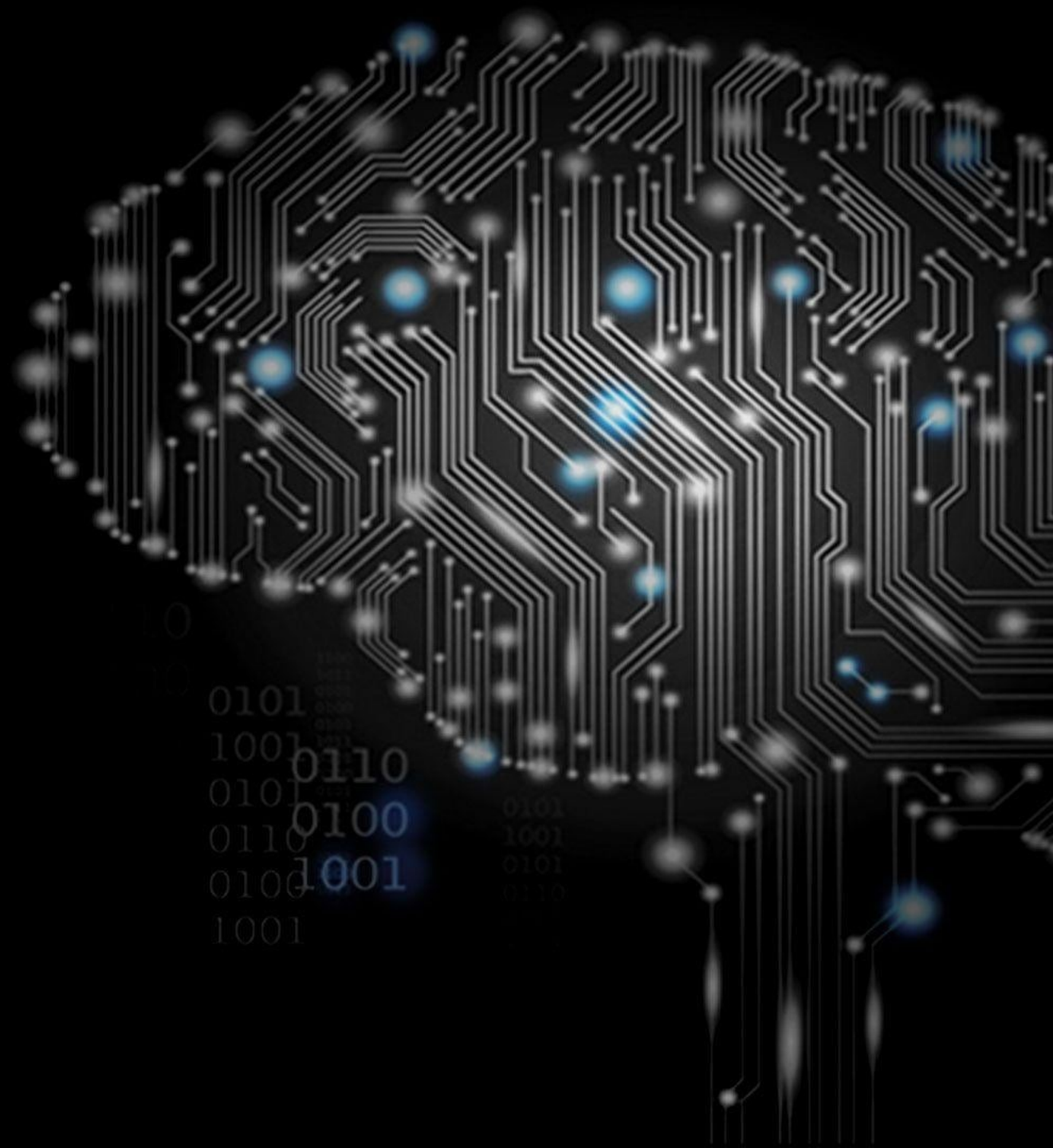

Introduction to Generative AI

6. Fine-Tuning and Model Adaptation

Module Topics

- Supervised Fine Tuning
- PEFT
- LoRA Fundamentals
- QLoRA Fundamentals
- Fine-Tuning Workflow
- Evaluation of Fine-Tuned Models

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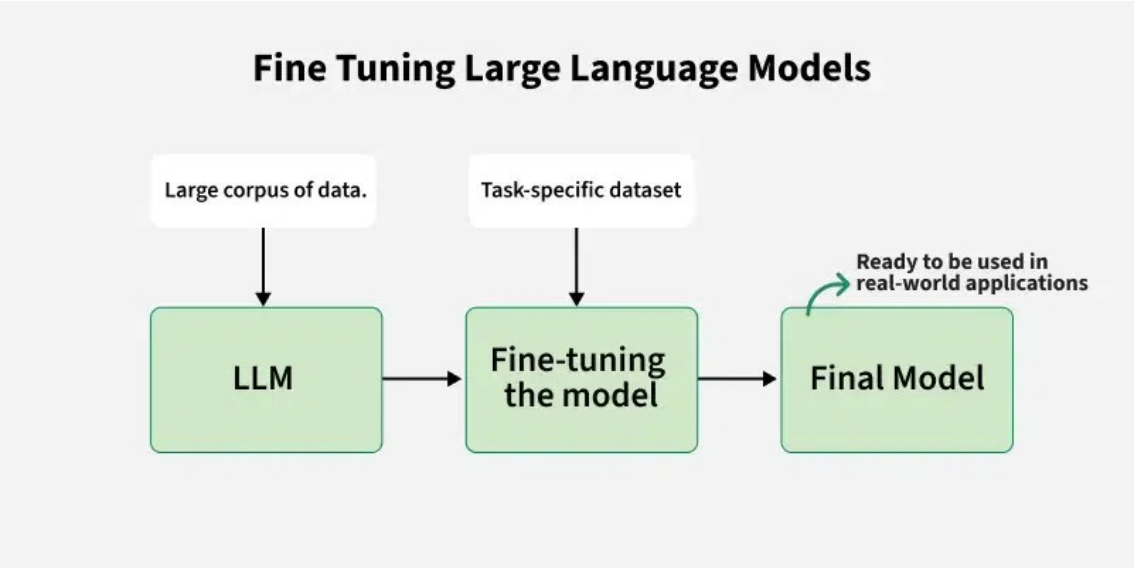
Fine Tuning

- Fine-tuning
  - The process of updating a pretrained model's parameters by using additional task-specific data to
    - Modify its behavior
    - Improve performance
    - Adapt it to a specific domain.
  - Formally,
    - Given a pretrained model with parameters θ
    - Fine-tuning performs: $\theta' = \theta + \Delta\theta$
    - θ = original pretrained weights
    - $\Delta\theta$ = learned adjustments from new training data
    - θ' = adapted model



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Fine Tuning

- Fine-tuning
  - Changes model weights
  - Requires training (backpropagation)
  - Usually uses labeled data
  - Produces a new model variant
  - Alters behavior persistently across all prompts
- Different from
  - Prompt engineering (no weight changes)
  - RAG (adds knowledge but no weight changes)

Method	Changes Model Weights?	Persistent Behavior Change?	Adds External Knowledge?
Prompting	No	No	No
RAG	No	No	Yes
Fine-Tuning	Yes	Yes	No

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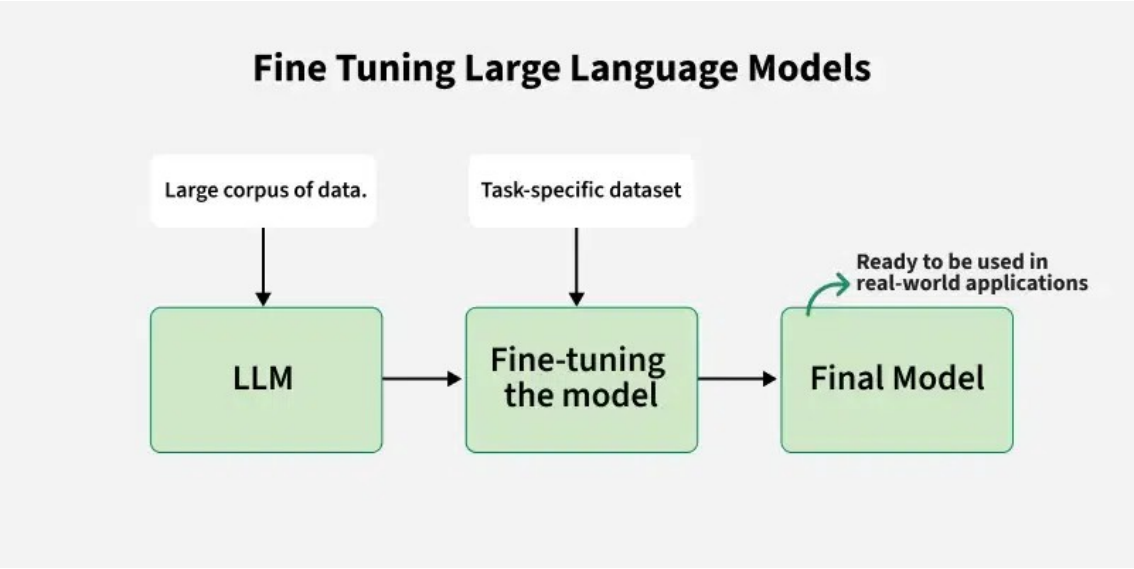
Fine Tuning

- When to fine tune
  - Should not start with fine tuning if the other options work because of the cost of fine-tuning
- Choice hierarchy
  - Prompt engineering
    - Cheapest
    - Fastest
    - No infrastructure changes
    - Works for many tasks



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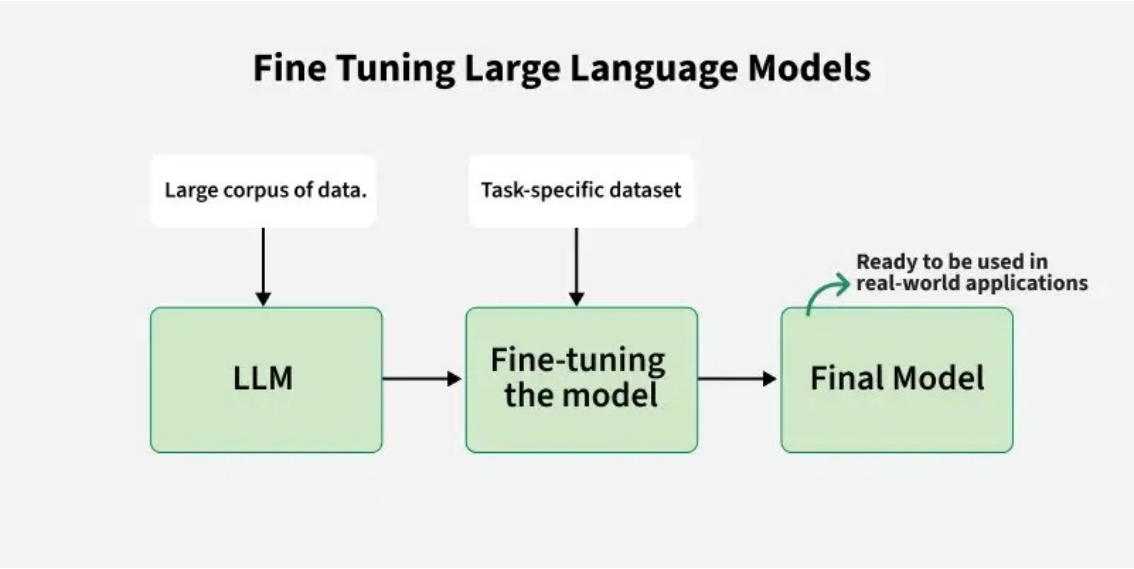
Fine Tuning

- Choice hierarchy
  - RAG (Retrieval-Augmented Generation)
    - Adds knowledge
    - Reduces hallucination
    - Still no weight changes
  - Fine-Tuning
    - Changes model behavior
    - Persistent adaptation
    - Higher cost and complexity



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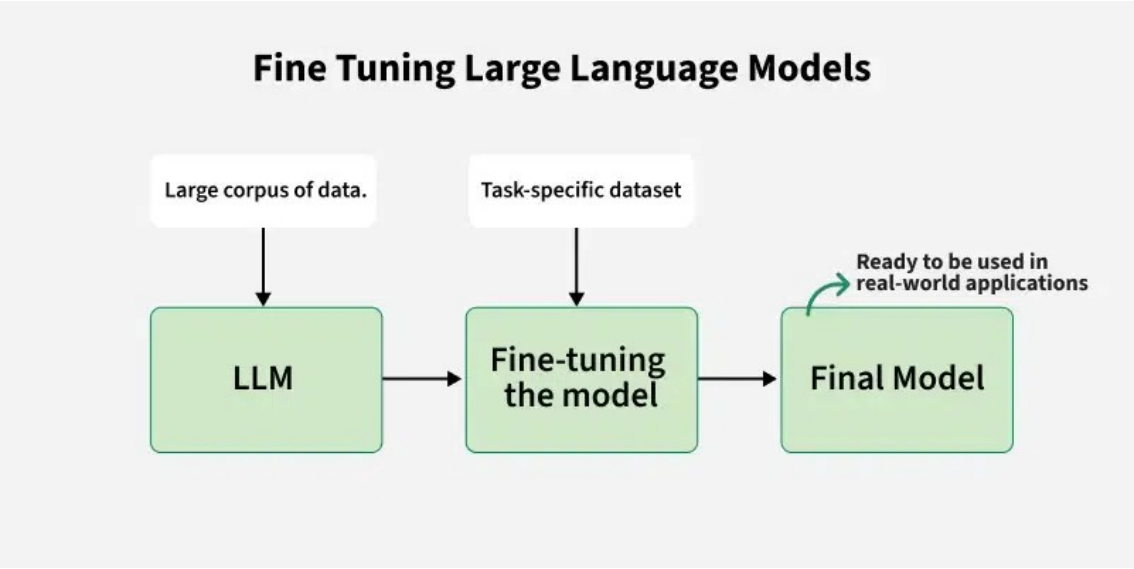
Fine Tuning

- Choice hierarchy
  - Use prompting when
    - Tasks are general-purpose
    - Behavior is acceptable with clear instructions
    - Tone inconsistencies are minor
    - Format reliability is not mission-critical
  - Use RAG when
    - The problem is missing knowledge
    - You need up-to-date data
    - Hallucinations come from knowledge gaps
    - The model's reasoning is fine but lacks facts
    - RAG does NOT change model behavior
    - It only augments context



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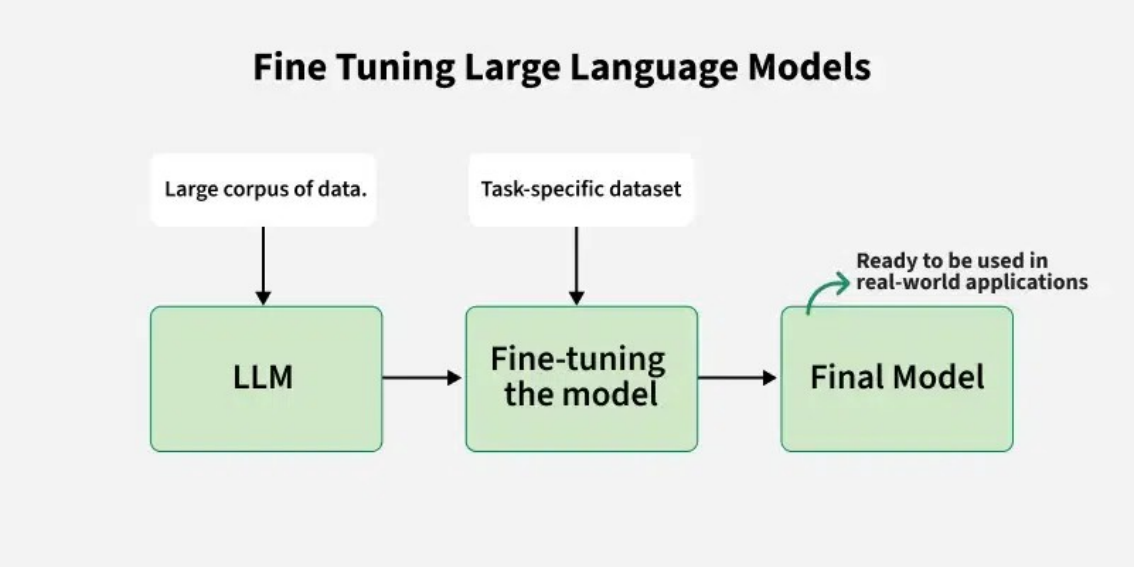
Fine Tuning

- Fine-tuning is justified when:
  - Behavioral inconsistency persists
    - Brand voice drifts
    - Tone varies
    - Structured output fails too often
  - Domain adaptation is needed
    - Legal reasoning
    - Medical terminology
    - Financial compliance language
  - Structured output must be Highly reliable
    - Classification pipelines
    - Extraction systems
    - Tool invocation formatting
  - Latency matters
    - Large prompts + RAG increase token cost
    - Fine-tuned model reduces prompt size



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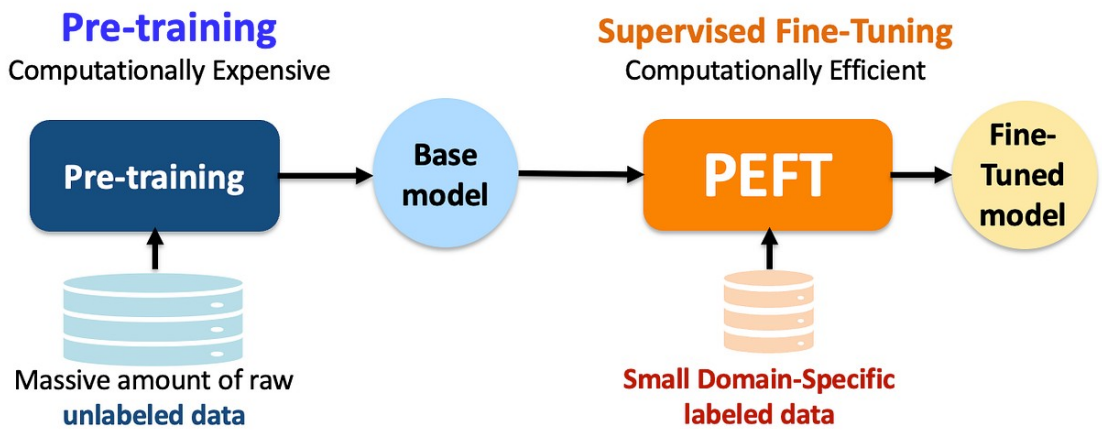
Supervised Fine-Tuning

- Train on labeled input-output pairs
- Objective
  - Minimize next-token prediction loss on desired outputs
  - Training goal for the model
    - *"For this type of input, respond like this"*
- Where SFT Works Well
  - Domain-specific Q&A
  - Classification tasks
  - Extraction tasks
  - Tone/style adaptation
  - Tool invocation formatting



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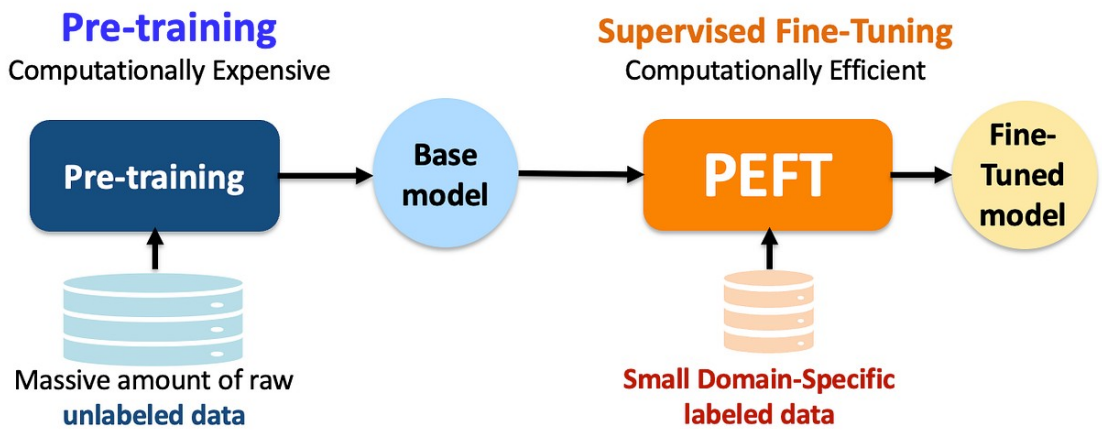
Supervised Fine-Tuning

- Training data requirements
  - Quality data more important than quantity
  - Important considerations:
    - Clean formatting
    - Consistent structure
    - No conflicting examples
    - No hallucinated labels
    - Balanced class distribution (for classification)
    - Poor data quality causes behavior degradation.



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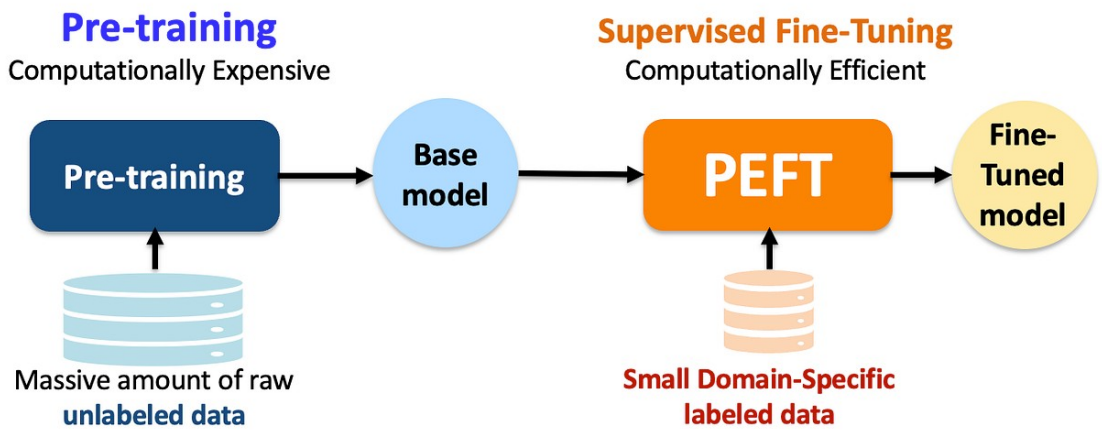
Supervised Fine-Tuning

- Full fine-tuning
  - Updates all parameters
  - High compute cost
  - High memory requirement
  - Risk of catastrophic forgetting
- Best suited for small models or research settings
  - Rarely used for large enterprise models



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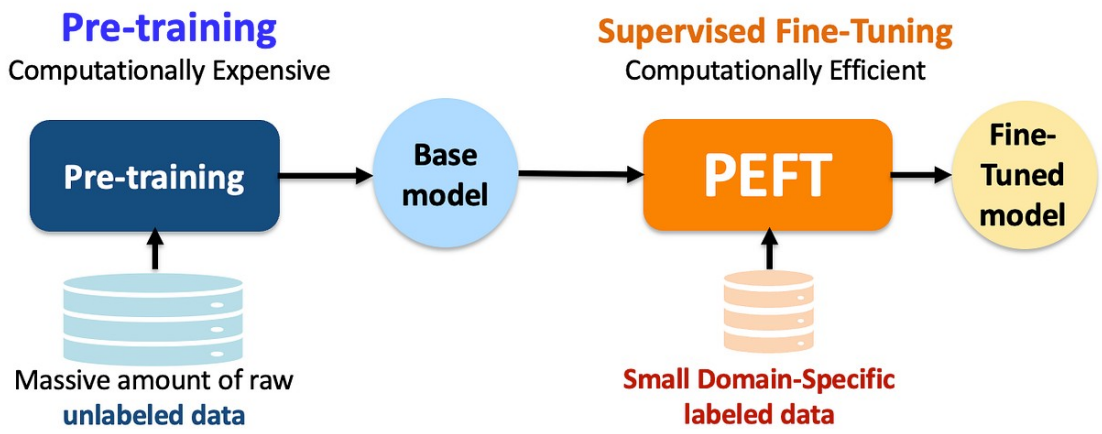
Supervised Fine-Tuning

- Parameter-Efficient Fine-Tuning
  - Modern standard
    - Freeze base model
    - Train small additional parameter layers
    - Lower compute
    - Lower risk
  - Dominates production fine-tuning workflows in 2026
- Why freeze the base model
  - Because updating billions of parameters
    - Requires massive GPU memory
    - Requires large datasets
    - Is computationally expensive
    - Risks damaging general knowledge
  - Freezing preserves the pretrained intelligence



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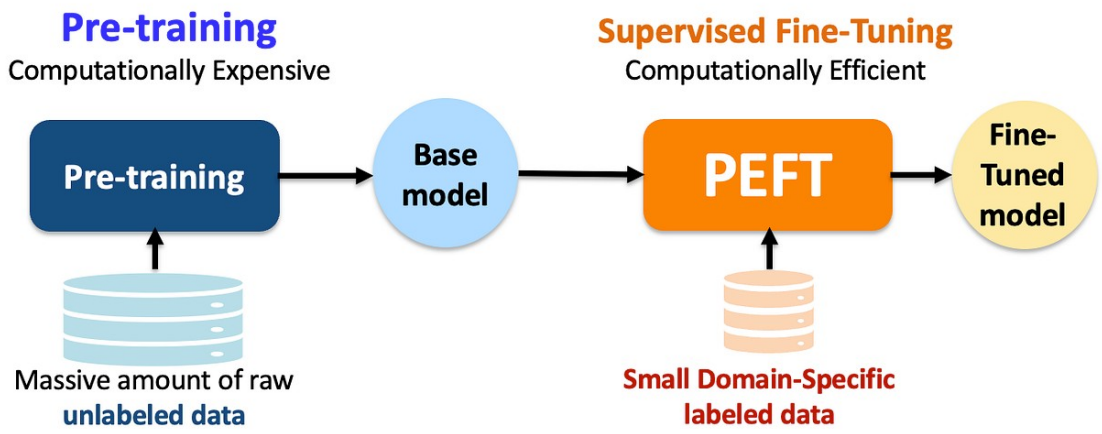
Supervised Fine-Tuning

- PEFT methods add small trainable modules
  - Injects them into specific transformer components
  - Keeps original weights untouched
  - These added components are:
    - LoRA adapters
  - Adapter layers
  - Prefix embeddings
  - Gating vectors (IA3)
- These are tiny compared to the base model



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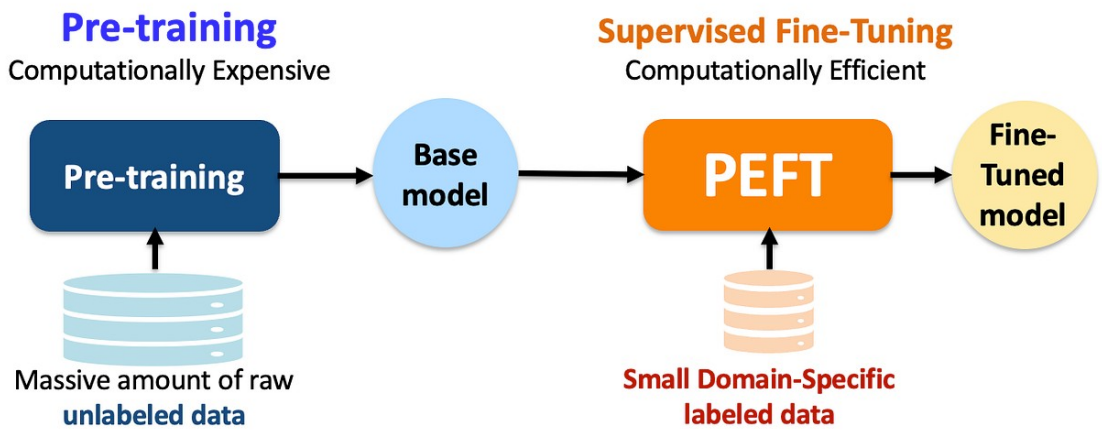
Supervised Fine-Tuning

- Imagine the base model is a frozen expert
  - Instead of retraining the expert,
  - PEFT attach “adjustment knobs” that
    - Slightly shift internal activations
    - Modify attention patterns
    - Bias outputs toward desired behavior
  - The core knowledge remains intact.
    - We’re adjusting behavior, not rebuilding intelligence



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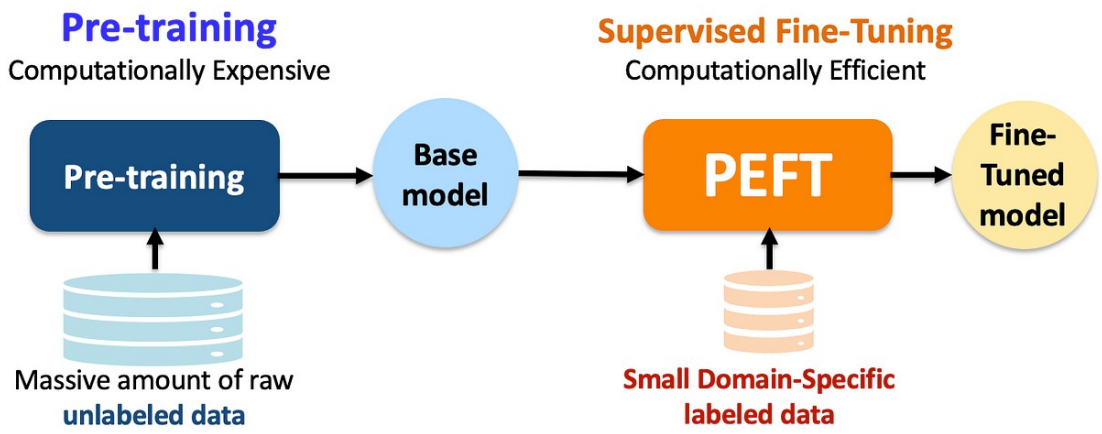
Supervised Fine-Tuning

- Full fine-tuning risks
- Catastrophic forgetting
  - Model may lose
    - General reasoning ability
    - Safety alignment
    - Broad knowledge
  - Because all weights shift
- Overfitting
  - With small domain dataset:
    - Entire model over-specializes
    - Generalization drops



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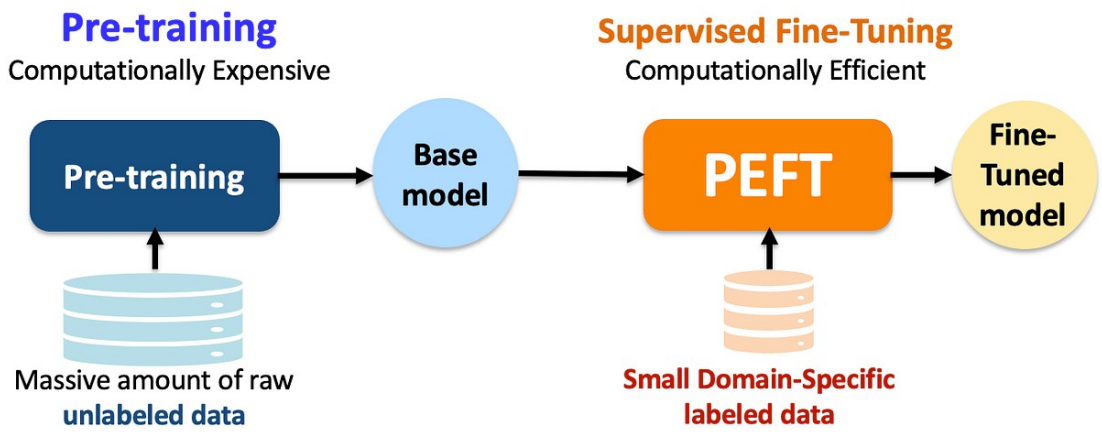
Supervised Fine-Tuning

- Full fine-tuning risks
- Irreversible degradation
  - Once full weights change:
    - Harder to revert
    - Larger checkpoint sizes



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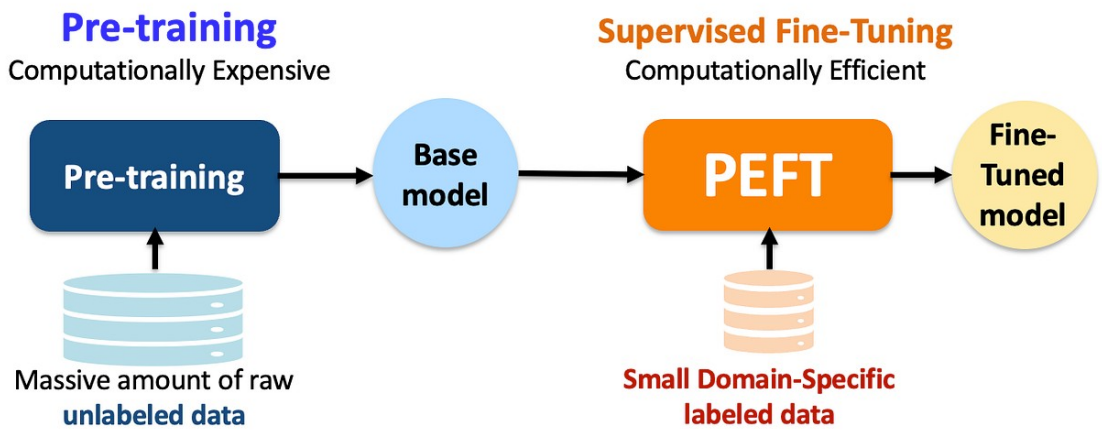
Supervised Fine-Tuning

- PEFT reduces these risks
  - Because
    - Base knowledge is frozen
    - Changes are modular
    - Adapters can be removed
    - Easy rollback
  - If adapter fails
    - Detach it
    - Base model remains intact
  - This modularity is critical in production.



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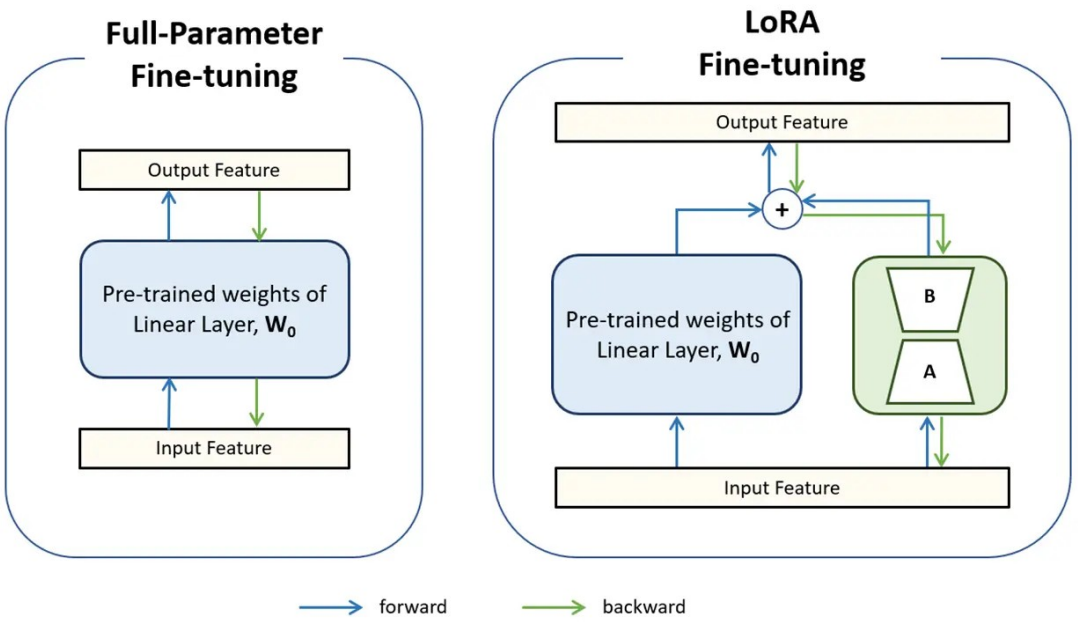
LoRA Fundamentals

- LoRA fundamentals
  - Most widely used PEFT method in modern LLM systems
  - As of 2026, LoRA is the default fine-tuning approach for most enterprise teams
- LoRA stands for
  - Low-Rank Adaptation
  - Instead of modifying the full weight matrix we transform W'=W+AB where
    - W is the original pretrained weight matrix (frozen)
    - A and B are small trainable matrices
    - Only A and B are updated



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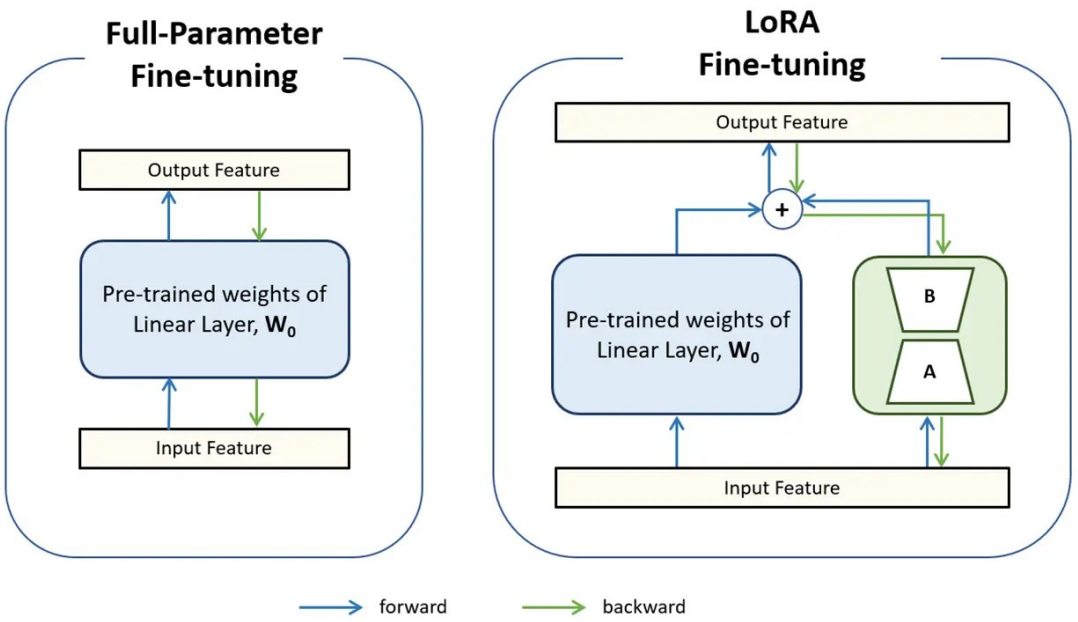
LoRA Fundamentals

- Why LoRA works
  - Large pretrained models already contain rich general knowledge
  - Most domain adaptation does NOT require rewriting all weights
  - It only requires
    - Small directional shifts
    - Slight reweighting of internal activations
    - Modifying attention patterns
  - LoRA captures those adjustments in low-rank form



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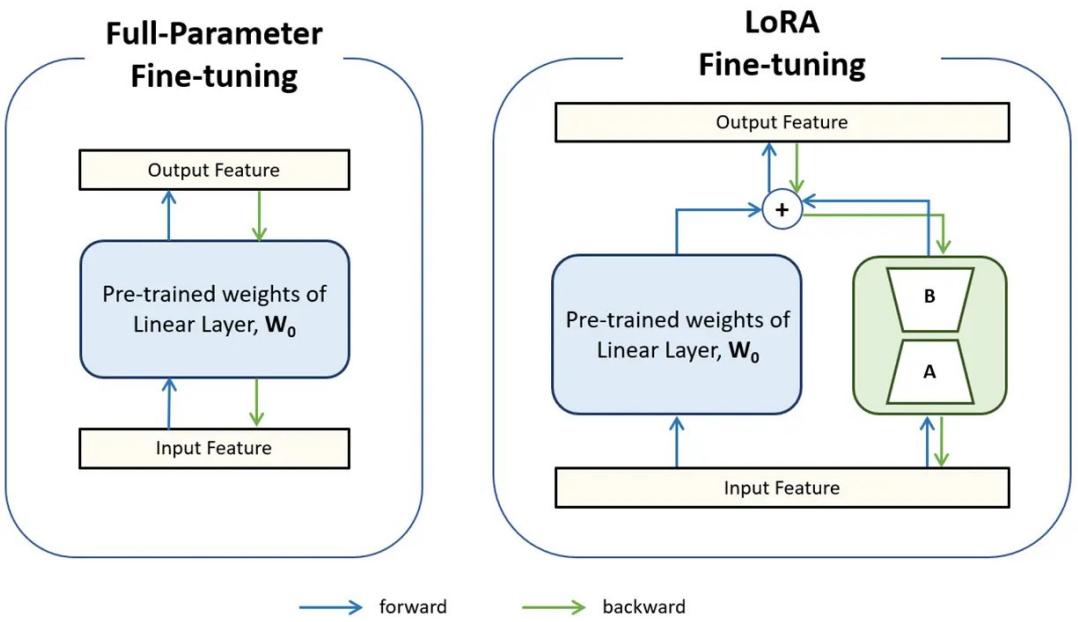
LoRA Fundamentals

- Advantages
  - Lower GPU Memory
  - Full fine-tuning requires
    - Storing gradients for billions of parameters
    - Storing optimizer states (often 2× parameters)
    - High memory bandwidth
  - LoRA requires
    - Gradients only for small matrices
    - Tiny optimizer state
    - Frozen base weights
    - This reduces memory usage dramatically.



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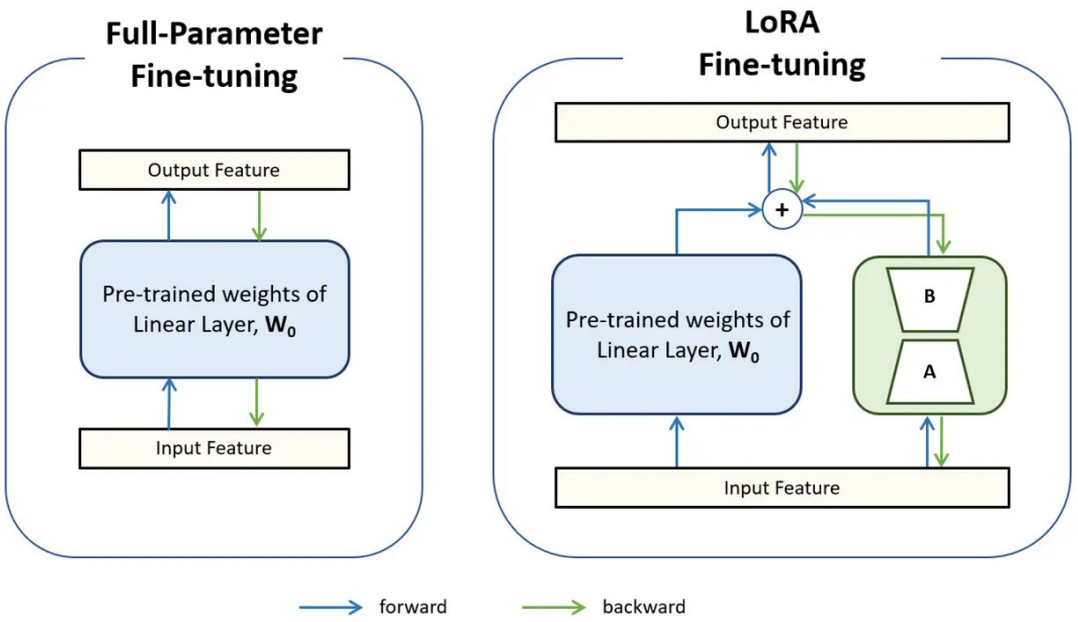
LoRA Fundamentals

- Advantages
  - Faster experimentation
  - Because
    - Fewer parameters are trained
    - Smaller memory footprint
    - Faster gradient computation
  - This means fine tuning can
    - Train faster
    - Iterate more
    - Run more experiments
    - Tune hyperparameters easily
  - This is critical in production environments



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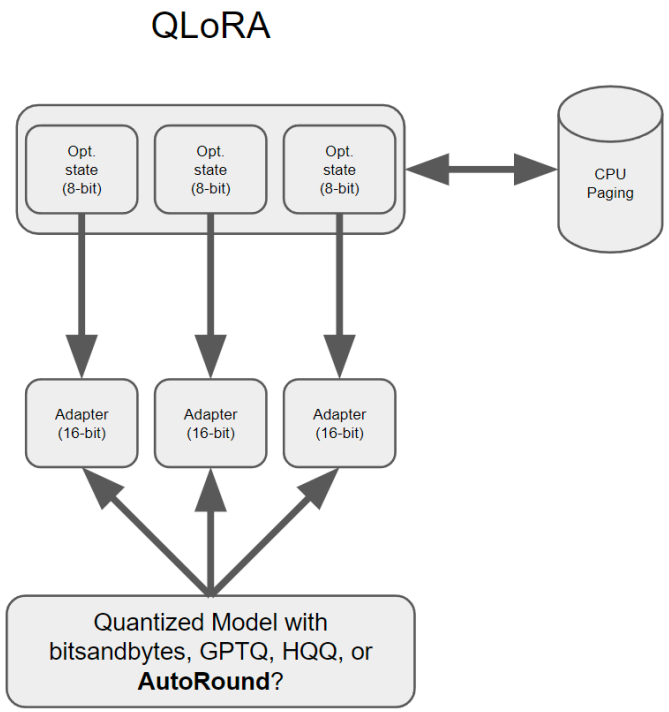
QLoRA Fundamentals

- QLoRA combines:
  - Quantization
  - LoRA adapters
- Quantize the base model
  - Instead of storing base weights in 16-bit or 32-bit precision
  - They are compressed to 4-bit precision
    - This reduces memory dramatically
    - The base model still remains frozen



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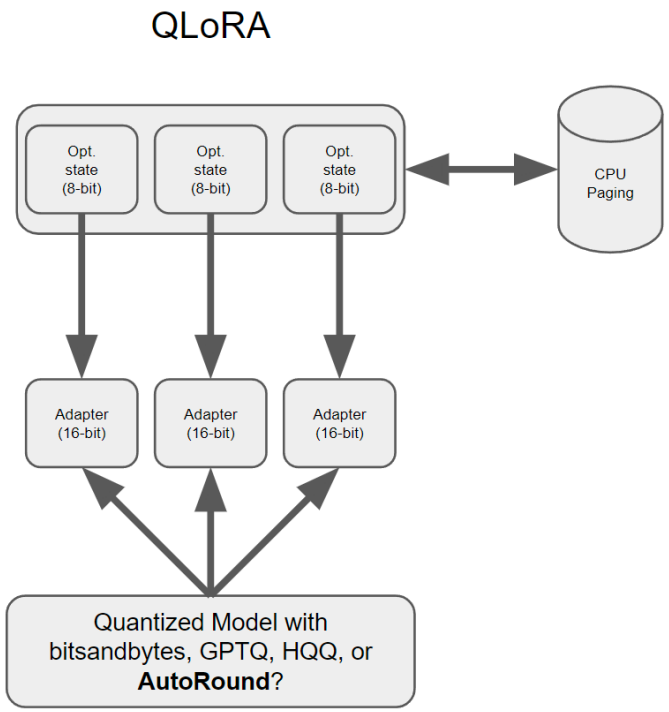
QLoRA Fundamentals

- Apply LoRA adapters
  - Even though the base model is quantized
    - LoRA adapters are trained in higher precision
    - Gradients flow through quantized weights
  - Only adapters are updated which means
    - Very large models can be fine tuned on modest hardware
- Substantially reduces costs of fine tuning



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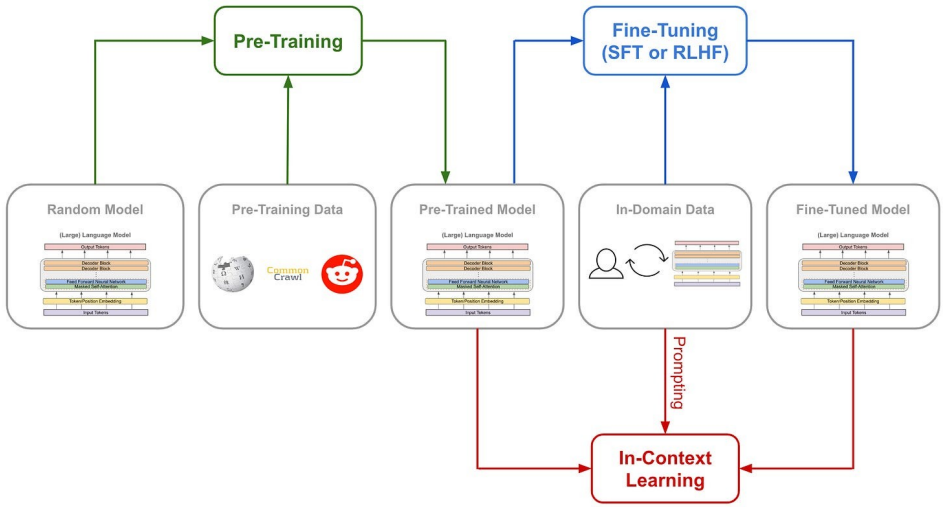
Fine Tuning Workflow

- Select base model
  - This is the most important strategic decision
    - You are not building intelligence from scratch
    - You are adapting an existing foundation.
  - Questions to Ask
    - Does the base model already perform well on my task?
    - Is it instruction-tuned?
    - Is it aligned and safe?
    - What is its size vs hardware constraints?
    - Is it open-weight or API-only?
  - A weak base model cannot be “fixed” by fine-tuning
  - A strong base model requires only small adaptation



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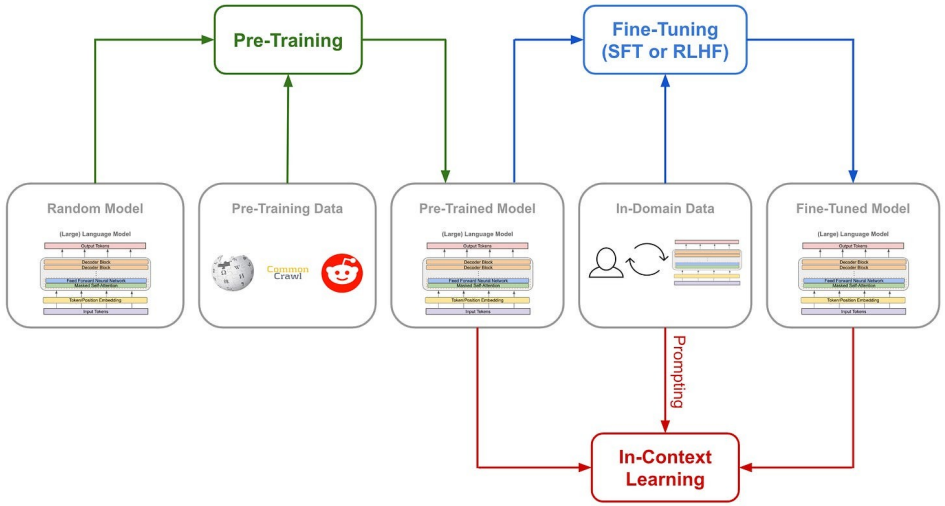
Fine Tuning Workflow

- Prepare dataset
  - Defines the behavior to be embedded
    - Fine-tuning is supervised learning
    - Your dataset is the specification
  - The model will learn patterns in the dataset
  - If the data is
    - Inconsistent then the model becomes inconsistent
    - Biased then the model amplifies bias
    - Noisy then the model degrades



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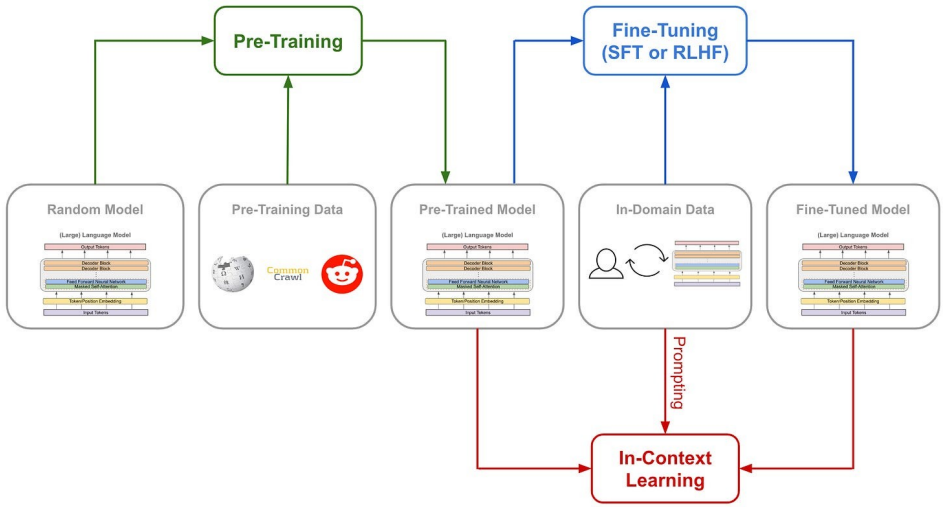
Fine Tuning Workflow

- Prepare dataset
  - A few thousand high-quality examples can outperform 100k noisy examples
  - The dataset must reflect
    - Desired tone
    - Desired structure
    - Desired correctness
    - Edge cases
    - Failure modes



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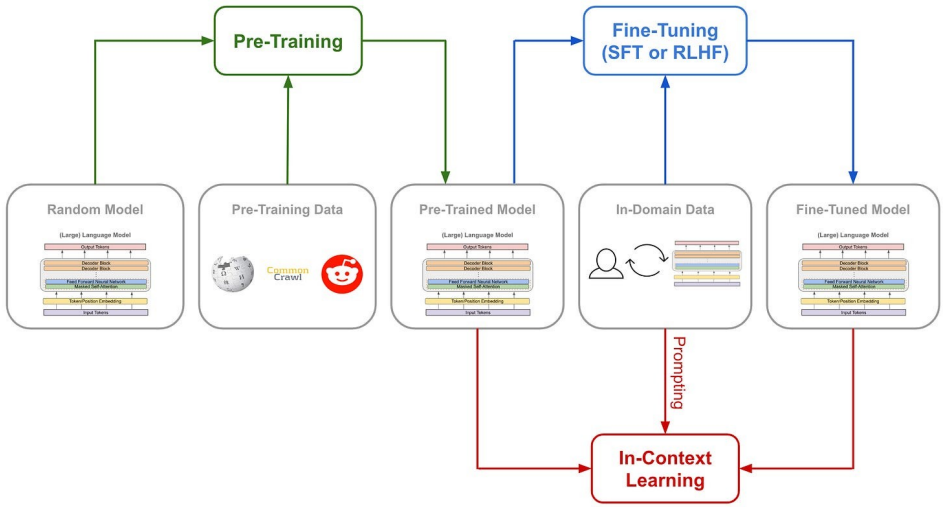
Fine Tuning Workflow

- Format training examples
  - The formatting determines what the model learns about structure
  - The model learns
    - Conversational structure
    - Role behavior
    - Output format patterns
  - If formatting is inconsistent, behavior becomes inconsistent



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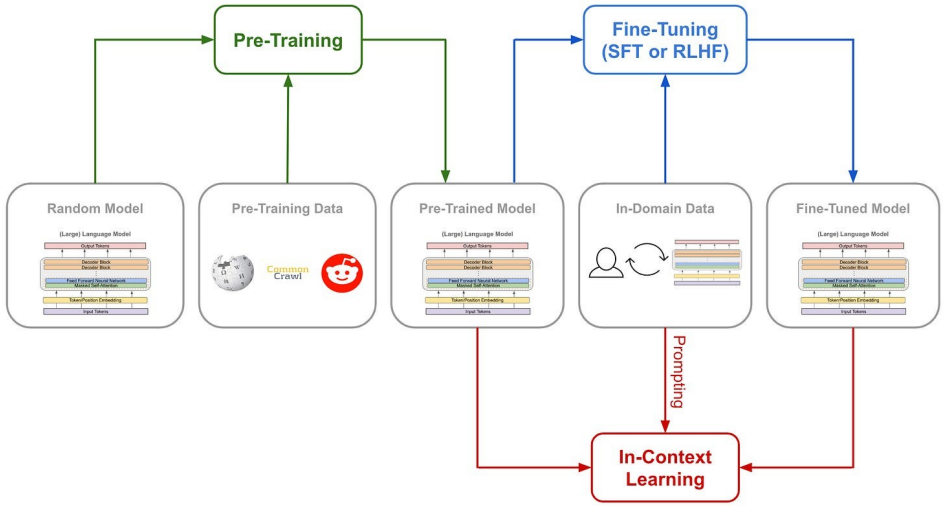
Fine Tuning Workflow

- Tokenization
  - Often overlooked conceptually
  - Tokenization determines how the model sees text
    - Words are broken into subword tokens
    - Special tokens define roles or separators
    - Sequence length matters
  - Important considerations
    - Are prompts truncated?
    - Are long examples clipped?
    - Are special tokens used correctly?
  - Fine-tuning cannot fix tokenization mistakes



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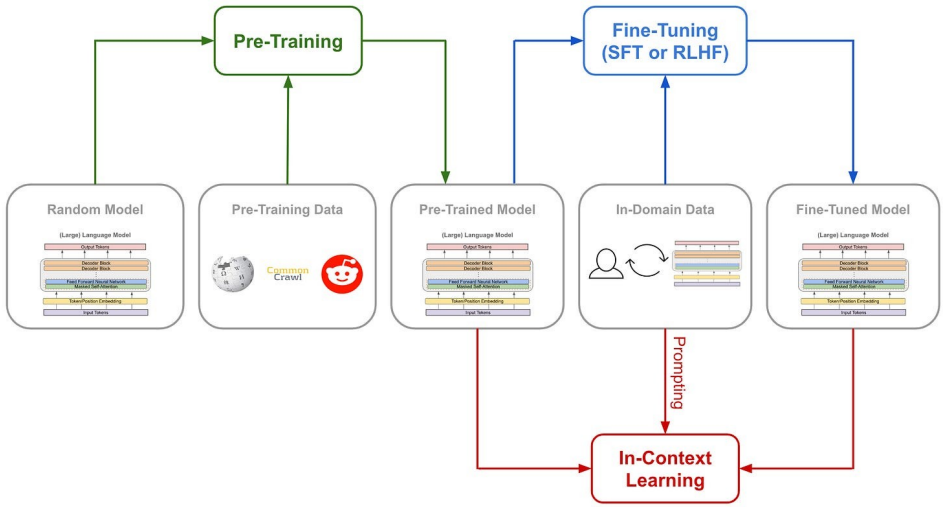
Fine Tuning Workflow

- Choose adaptation method
  - Full fine-tuning or PEFT
  - Modern best practice (2026)
    - Use PEFT unless strong reason not to.
    - Freeze base model.
    - Train lightweight adapters.
  - Reasons
    - Lower compute
    - Lower risk
    - Faster iteration
    - Modular deployment
  - Full fine-tuning is rarely needed for large models



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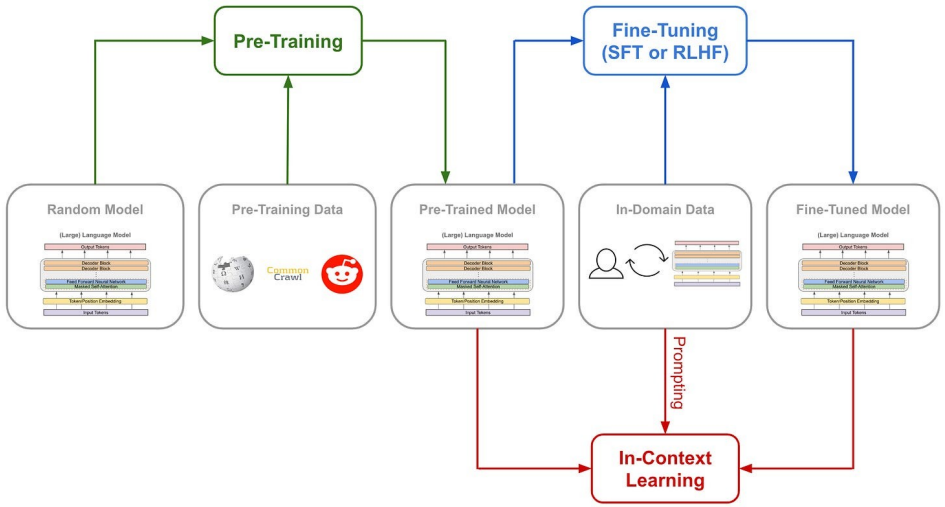
Fine Tuning Workflow

- Training configuration
  - How aggressively do we update parameters?
  - Important factors
    - Learning rate
    - Batch size
    - Number of epochs
    - Rank (for LoRA)
    - Regularization
  - Too aggressive
    - Overfitting
    - Catastrophic forgetting
  - Too weak
    - No improvement
  - The goal is controlled behavioral shift



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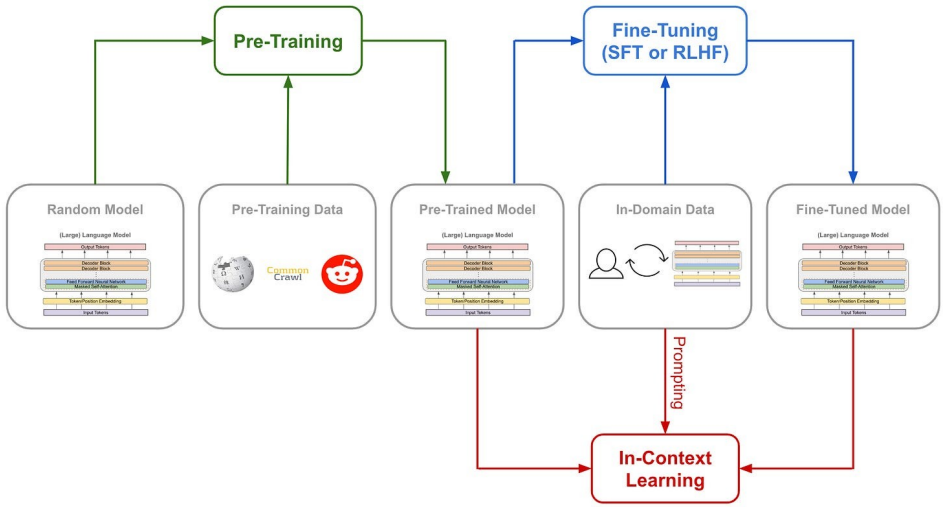
Fine Tuning Workflow

- Evaluation
  - Determines whether adaptation helped or harmed
  - We compare base model vs fine-tuned model
    - Target task
    - General tasks
    - Safety behaviors
    - Hallucination rates
  - Fine-tuning can improve one metric while degrading another
  - Evaluation must be multi-dimensional



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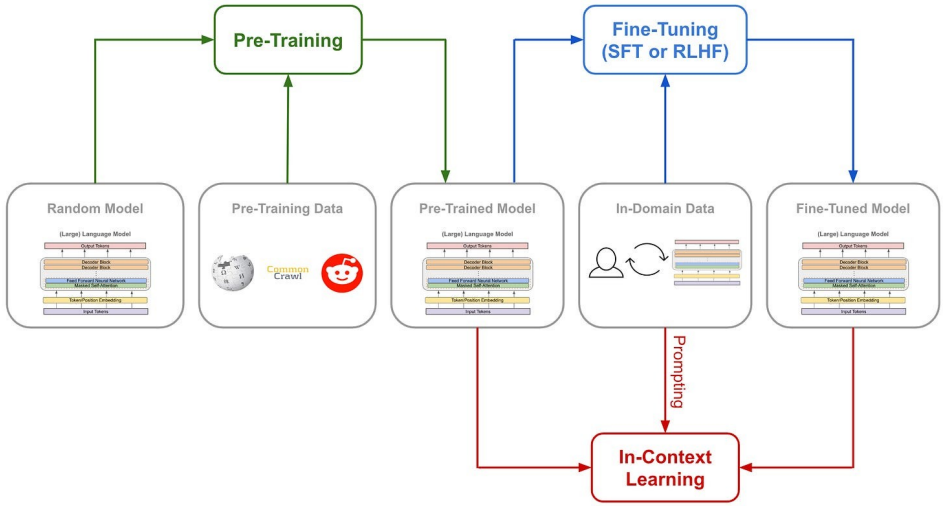
Fine Tuning Workflow

- Deployment
  - Fine-tuned model becomes a production artifact
  - Key concerns
    - Versioning
    - Rollback capability
    - Monitoring performance drift
    - Cost of inference
    - Latency impact



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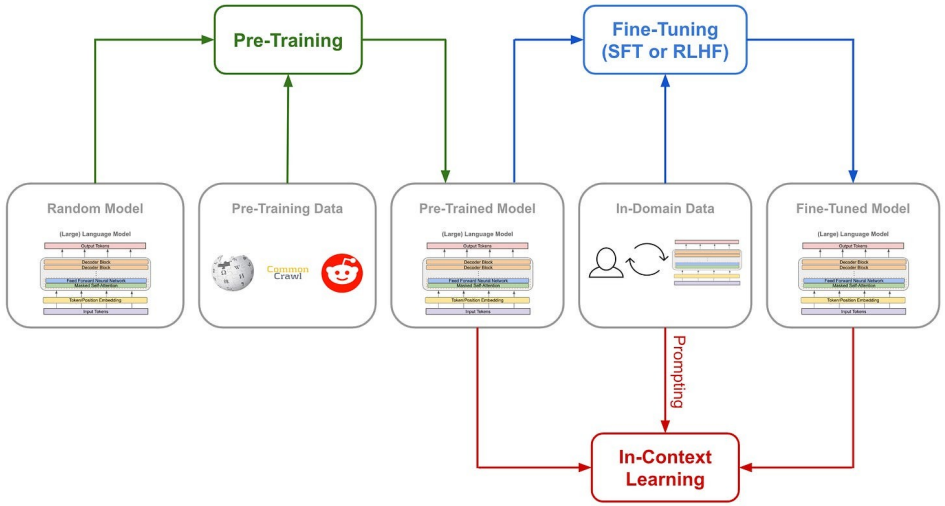
Evaluation

- Validation datasets
  - Separate dataset from training
  - Used to detect
    - Overfitting
    - Memorization
    - Instability
  - If performance drops outside training data,you have over-specialized



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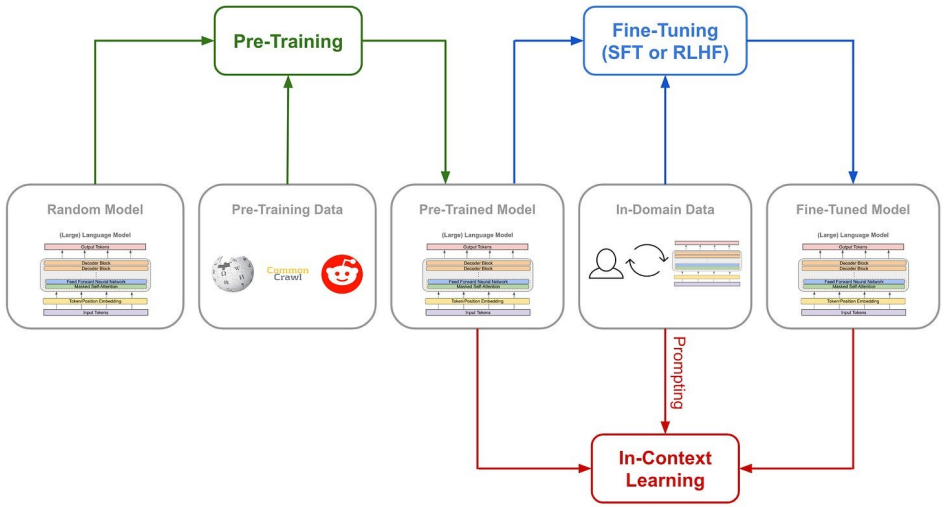
Evaluation

- Validation datasets
  - Overfitting detection
    - Training accuracy very high
    - Validation accuracy lower
    - Model repeats training patterns verbatim
    - Reduced creativity or flexibility
  - Fine-tuned model should generalize within domain not memorize



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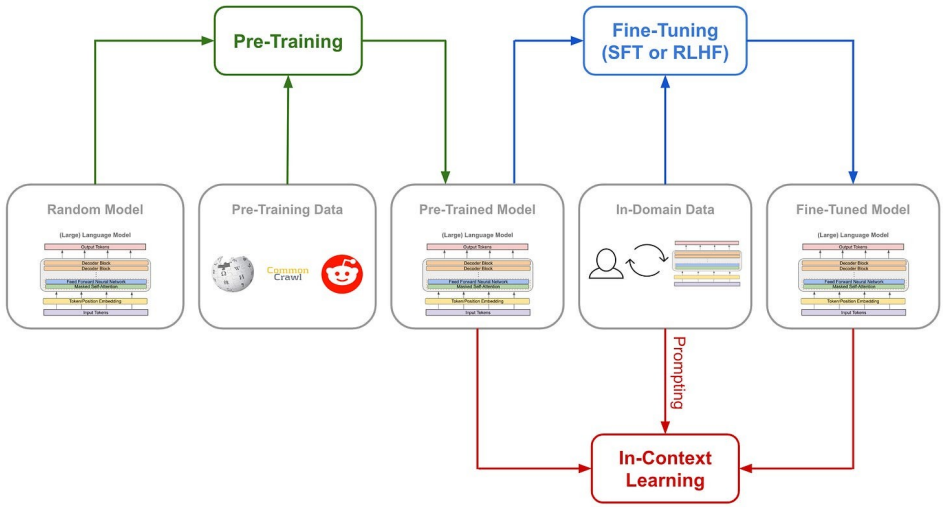
Evaluation

- Generalization checks
  - Test on
    - Slightly different phrasing
    - New examples
    - Edge cases
    - Out-of-distribution inputs
  - Question
    - Did we improve the intended task without damaging broader reasoning ability?



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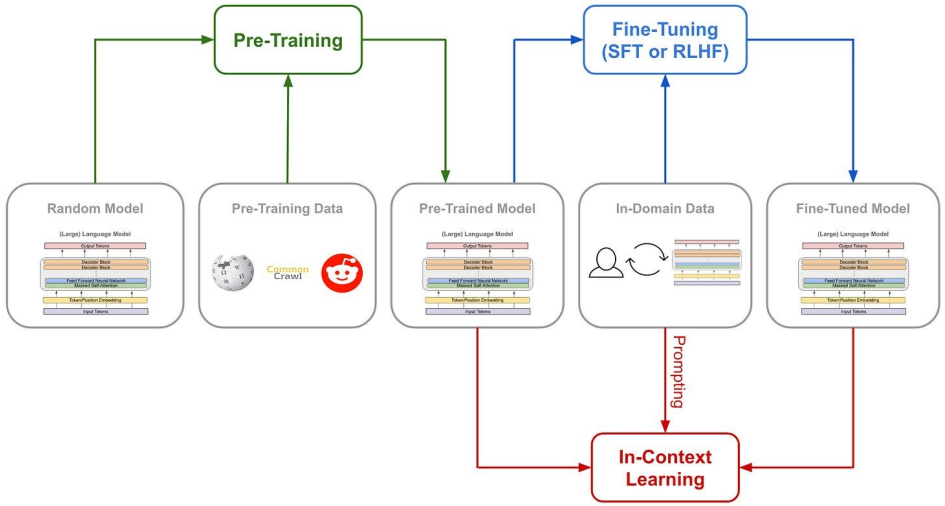
Evaluation

- Hallucination comparison
  - Fine-tuning may
    - Reduce hallucination (good)
    - Increase hallucination in narrow domain (bad)
  - Measure
    - Unsupported claims
    - Incorrect facts
    - Fabricated details
    - Compare base vs fine-tuned.



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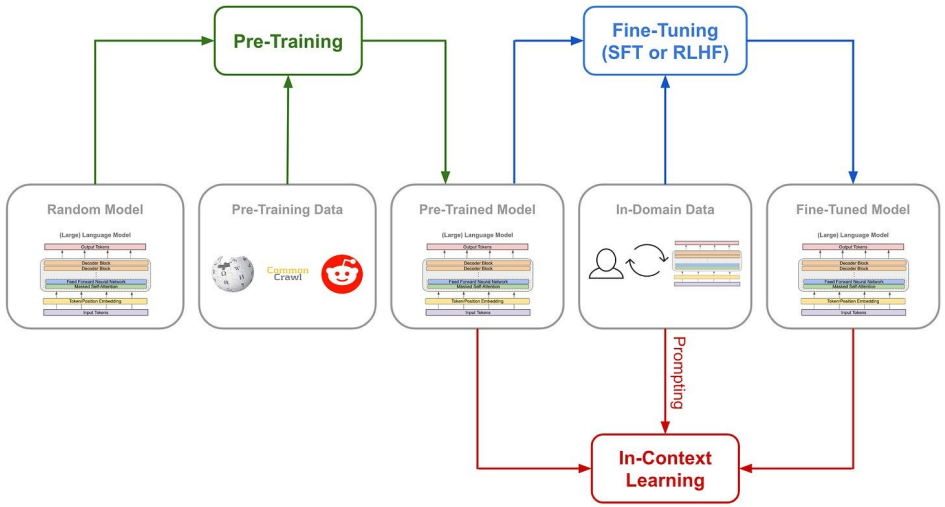
Evaluation

- Regression testing
  - Critical in production
  - Ensure that
    - Previously working tasks still work.
    - Safety behavior is intact
    - Structured output compliance remains stable.
  - Fine-tuning can unintentionally degrade capabilities
  - Regression testing protects against silent failure



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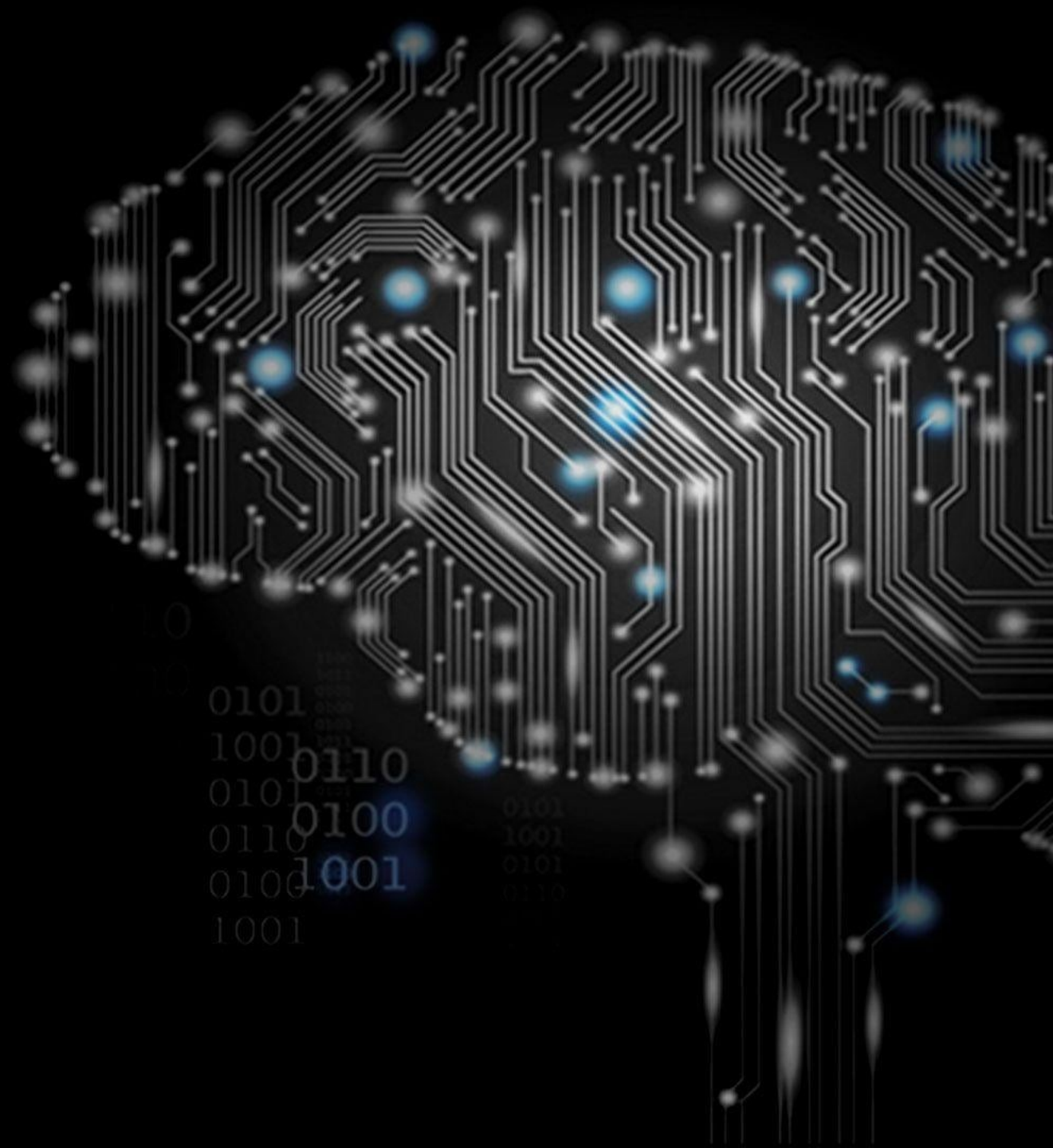

Questions?